

Shriman Oppilamani

 github.com/shrimanpro |  linkedin.com/in/shriman-oppilamani |  shriman.pro | 
shriman.oppilamani@gmail.com |  (510)-766-6959

EDUCATION

University of California - Los Angeles

Expected June 2028

Master of Science in Computer Science

Arizona State University

May 2026

Bachelor of Science in Computer Science, GPA 4.10/4

SKILLS

Languages: C, C++, Rust, Python, Java, SQL, JavaScript, Bash

Technologies: Linux, MySQL, MariaDB, FastAPI, React, Git, Systemd, Unix APIs

EXPERIENCE

Software Engineering Capstone Developer (VaristySportsShow)

Aug 2025 - May 2026

Linux, Python, Bash

- Engineered a Flask backend processing high-throughput media, using memory-efficient file chunking to stream I/O directly to Backblaze B2.
- Developed secure REST APIs for gated subscriber access, interfacing with MySQL to validate sessions and manage media metadata.
- Collaborated in an agile team to architect and deliver a production-ready system for a commercial client.

PROJECTS

Melody OS (Bare-Metal Audio Operating System)

Jan 2026 - Present

Rust, AArch64, Hardware Drivers

- Architecting a bare-metal kernel from scratch in Rust for the Raspberry Pi, designed specifically as a low-latency, lossless audio sequencer.
- Developing custom device drivers to directly interface with I2S audio peripherals and hardware DACs, bypassing traditional OS scheduler overhead.
- Implementing memory-safe interrupt handling and lock-free concurrency utilizing Rust's ownership model to ensure uninterrupted, high-fidelity audio buffering.

Low-Level Text Editor

Nov 2025 - Present

C, Make, Unix Systems

- Built a dependency-free C text editor, directly utilizing POSIX APIs and termios to handle raw terminal input.
- Engineered a custom TUI rendering engine with ANSI escape sequences, avoiding high-level libraries like ncurses.
- Managed dynamic memory allocation for unbounded row buffers and file I/O, rigorously preventing memory leaks.

Custom Pseudocode Compiler – Solo Class Project

April 2025 - May 2025

C++

- Architected a C++ compiler frontend from scratch with a custom lexical scanner and recursive descent parser.
- Constructed a dynamic Abstract Syntax Tree (AST) to efficiently evaluate complex control flows and operations.
- Enforced strict manual memory management within the execution engine for optimized, memory-safe AST traversal.

Linux Kernel and Hardware Optimization

Dec 2024 - Present

Arch Linux, Bash

- Patched the Linux kernel to override ACPI thermal polling, resolving a severe hardware-level CPU throttle (locked at 200MHz).
- Debugged proprietary network driver stacks and wpa-supPLICANT to fix persistent WPA2-Enterprise auth failures.
- Configured advanced power management states and systemd daemons to stabilize non-native hardware operation.